

Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

Recall that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

The foliage of your bonsai tree is a perfect sphere. The radius of the foliage is increasing at a rate of 0.5 millimeters per day. How fast is the volume of the sphere increasing when the radius is 100 millimeters?

Exercise 1.

An oil tanker is anchored 20 kilometers west of the Rose Garden. A windsurfer is 80 kilometers north of the Rose Garden, moving north at a speed of 60 kilometers per hour. At this moment, how fast is the distance between the oil tanker and the windsurfer increasing?

Exercise 2.

A 20 meter long akebono tree was bent in a rainstorm and now makes an angle of $\frac{\pi}{3}$ radians with the ground. A cherry blossom is sliding down the trunk at a speed of 5 centimeters per second. When the flower is halfway down the tree, how fast is the distance from the flower to the ground decreasing?